nonnegative capacity $c(u, v) \geq 0$ and a cost $a(u, v)$. As in the multicommodity-flow problem, there are k different commodities, K_1, K_2, … , K_k, with commodity i specified by the triple $K_i = (s_i, t_i, d_i)$. We define the flow f_i for commodity i and the aggregate flow f_{uv} on edge (u, v) as in the multicommodity-flow problem. A feasible flow is one in which the aggregate flow on each edge (u, v) is no more than the capacity of edge (u, v). The cost of a flow is $\sum_{u,v \in E} a(u, v) \cdot f_{uv}$, and the goal is to find the feasible flow of minimum cost. Express this problem as a linear program.

29.3 Duality

We will now introduce a powerful concept called ***linear-programming duality***. In general, given a maximization problem, duality allows you to formulate a related minimization problem that has the same objective value. The idea of duality is actually more general than linear programming, but we restrict our attention to linear programming in this section.

Duality enables us to prove that a solution is indeed optimal. We saw an example of duality in Chapter 24 with Theorem 24.6, the max-flow min-cut theorem. Suppose that, given an instance of a maximum-flow problem, you find a flow f with value $|f|$. How do you know whether f is a maximum flow? By the max-flow min-cut theorem, if you can find a cut whose value is also $|f|$, then you have verified that f is indeed a maximum flow. This relationship provides an example of duality: given a maximization problem, define a related minimization problem such that the two problems have the same optimal objective values.

Given a linear program in standard form in which the objective is to maximize, let's see how to formulate a ***dual*** linear program in which the objective is to minimize and whose optimal value is identical to that of the original linear program. When referring to dual linear programs, we call the original linear program the ***primal***.

Given the primal linear program

$$\text{maximize} \quad \sum_{j=1}^{n} c_j x_j \tag{29.31}$$

subject to

$$\sum_{j=1}^{n} a_{ij} x_j \leq b_i \quad \text{for } i = 1, 2, \ldots, m \tag{29.32}$$
$$x_j \geq 0 \quad \text{for } j = 1, 2, \ldots, n \ , \tag{29.33}$$

its dual is

$$\text{minimize} \quad \sum_{i=1}^{m} b_i y_i \tag{29.34}$$

subject to

$$\sum_{i=1}^{m} a_{ij} y_i \geq c_j \quad \text{for } j = 1, 2, \ldots, n \tag{29.35}$$
$$y_i \geq 0 \quad \text{for } i = 1, 2, \ldots, m \ . \tag{29.36}$$

Mechanically, to form the dual, change the maximization to a minimization, exchange the roles of coefficients on the right-hand sides and in the objective function, and replace each ≤ by ≥. Each of the *m* constraints in the primal corresponds to a variable y_i in the dual. Likewise, each of the *n* constraints in the dual corresponds to a variable x_j in the primal. For example, consider the following primal linear program:

$$\text{maximize} \quad 3x_1 + x_2 + 4x_3 \tag{29.37}$$

subject to

$$x_1 + x_2 + 3x_3 \leq 30 \tag{29.38}$$
$$2x_1 + 2x_2 + 5x_3 \leq 24 \tag{29.39}$$
$$4x_1 + x_2 + 2x_3 \leq 36 \tag{29.40}$$
$$x_1, x_2, x_3 \geq 0 \ . \tag{29.41}$$

Its dual is

$$\text{minimize} \quad 30y_1 + 24y_2 + 36y_3 \tag{29.42}$$

subject to

$$y_1 + 2y_2 + 4y_3 \geq 3 \tag{29.43}$$
$$y_1 + 2y_2 + y_3 \geq 1 \tag{29.44}$$
$$3y_1 + 5y_2 + 2y_3 \geq 4 \tag{29.45}$$
$$y_1, y_2, y_3 \geq 0 \ . \tag{29.46}$$

Although forming the dual can be considered a mechanical operation, there is an intuitive explanation. Consider the primal maximization problem (29.37)–(29.41). Each constraint gives an upper bound on the objective function. In addition, if you take one or more constraints and add together nonnegative multiples of them, you get a valid constraint. For example, you can add constraints (29.38) and (29.39) to obtain the constraint $3x_1 + 3x_2 + 8x_3 \leq 54$. Any feasible solution to the primal must satisfy this new constraint, but there is something else interesting about it. Comparing this new constraint to the objective function (29.37), you can see that for each variable, the corresponding coefficient is at least as large as the coefficient in the objective function. Thus, since the variables x_1, x_2 and x_3 are nonnegative, we have that

$$3x_1 + x_2 + 4x_3 \leq 3x_1 + 3x_2 + 8x_3 \leq 54,$$

and so the solution value to the primal is at most 54. In other words, adding these two constraints together has generated an upper bound on the objective value.

In general, for any nonnegative multipliers y_1, y_2, and y_3, you can generate a constraint

$$y_1(x_1+x_2+3x_3)+y_2(2x_1+2x_2+5x_3)+y_3(4x_1+x_2+2x_3) \leq 30y_1+24y_2+36y_3$$

from the primal constraints or, by distributing and regrouping,

$$(y_1+2y_2+4y_3)x_1+(y_1+2y_2+y_3)x_2+(3y_1+5y_2+2y_3)x_3 \leq 30y_1+24y_2+36y_3.$$

Now, as long as this constraint has coefficients of x_1, x_2, and x_3 that are at least their objective-function coefficients, it is a valid upper bound. That is, as long as

$$
\begin{aligned}
y_1 + 2y_2 + 4y_3 &\geq 3, \\
y_1 + 2y_2 + y_3 &\geq 1, \\
3y_1 + 5y_2 + 2y_3 &\geq 4,
\end{aligned}
$$

you have a valid upper bound of $30y_1+24y_2+36y_3$. The multipliers y_1, y_2, and y_3 must be nonnegative, because otherwise you cannot combine the inequalities. Of course, you would like the upper bound to be as small as possible, and so you want to choose y to minimize $30y_1 + 24y_2 + 36y_3$. Observe that we have just described the dual linear program as the problem of finding the smallest possible upper bound on the primal.

We'll formalize this idea and show in Theorem 29.4 that, if the linear program and its dual are feasible and bounded, then the optimal value of the dual linear program is always equal to the optimal value of the primal linear program. We begin by demonstrating ***weak duality***, which states that any feasible solution to the primal linear program has a value no greater than that of any feasible solution to the dual linear program.

Lemma 29.1 (Weak linear-programming duality)

Let $\bar{x}$ be any feasible solution to the primal linear program in (29.31)–(29.33), and let $\bar{y}$ be any feasible solution to its dual linear program in (29.34)–(29.36). Then

$$
\sum_{j=1}^{n} c_j \bar{x}_j \leq \sum_{i=1}^{m} b_i \bar{y}_i \; .
$$

Proof We have

$$
\begin{aligned}
\sum_{j=1}^{n} c_j \bar{x}_j &\leq \sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_{ij} \bar{y}_i \right) \bar{x}_j && \text{(by inequalities (29.35))} \\
&= \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} \bar{x}_j \right) \bar{y}_i \\
&\leq \sum_{i=1}^{m} b_i \bar{y}_i && \text{(by inequalities (29.32))} \; .
\end{aligned}
$$

■

Corollary 29.2

Let $\bar{x}$ be a feasible solution to the primal linear program in (29.31)–(29.33), and let $\bar{y}$ be a feasible solution to its dual linear program in (29.34)–(29.36). If

$$\sum_{j=1}^{n} c_j \bar{x}_j = \sum_{i=1}^{m} b_i \bar{y}_i \,,$$

then $\bar{x}$ and $\bar{y}$ are optimal solutions to the primal and dual linear programs, respectively.

Proof By Lemma 29.1, the objective value of a feasible solution to the primal cannot exceed that of a feasible solution to the dual. The primal linear program is a maximization problem and the dual is a minimization problem. Thus, if feasible solutions $\bar{x}$ and $\bar{y}$ have the same objective value, neither can be improved.

■

We now show that, at optimality, the primal and dual objective values are indeed equal. To prove linear programming duality, we will require one lemma from linear algebra, known as Farkas's lemma, the proof of which Problem 29-4 asks you to provide. Farkas's lemma can take several forms, each of which is about when a set of linear equalities has a solution. In stating the lemma, we use $m + 1$ as a dimension because it matches our use below.

Lemma 29.3 (Farkas's lemma)

Given $M \in \mathbb{R}^{(m+1)\times n}$ and $g \in \mathbb{R}^{m+1}$, exactly one of the following statements is true:

1. There exists $v \in \mathbb{R}^n$ such that $Mv \le g$,
2. There exists $w \in \mathbb{R}^{m+1}$ such that $w \ge 0$, $w^{\mathrm{T}}M = 0$ (an n-vector of all zeros), and $w^{\mathrm{T}}g < 0$.

■

Theorem 29.4 (Linear-programming duality)

Given the primal linear program in (29.31)–(29.33) and its corresponding dual in (29.34)–(29.36), if both are feasible and bounded, then for optimal solutions x^* and y^*, we have $c^{\mathrm{T}}x^* = b^{\mathrm{T}}y^*$.

Proof Let $\mu = b^{\mathrm{T}}y^*$ be the optimal value of the dual linear program given in (29.34)–(29.36). Consider an augmented set of primal constraints in which we add a constraint to (29.31)–(29.33) that the objective value is at least μ. We write out this ***augmented primal*** as

$$Ax \le b \,, \tag{29.47}$$
$$c^{\mathrm{T}}x \ge \mu \,. \tag{29.48}$$

We can multiply (29.48) through by −1 and rewrite (29.47)–(29.48) as

$$\begin{pmatrix} A \\ -c^{\mathrm{T}} \end{pmatrix} x \le \begin{pmatrix} b \\ -\mu \end{pmatrix} . \tag{29.49}$$

Here, $\begin{pmatrix} A \\ -c^{\mathrm{T}} \end{pmatrix}$ denotes an $(m+1)\times n$ matrix, x is an n-vector, and $\begin{pmatrix} b \\ -\mu \end{pmatrix}$ denotes an $(m+1)$-vector.

We claim that if there is a feasible solution $\bar{x}$ to the augmented primal, then the theorem is proved. To establish this claim, observe that $\bar{x}$ is also a feasible solution to the original primal and that it has objective value at least μ. We can then apply Lemma 29.1, which states that the objective value of the primal is at most μ, to complete the proof of the theorem.

It therefore remains to show that the augmented primal has a feasible solution. Suppose, for the purpose of contradiction, that the augmented primal is infeasible, which means that there is no $v \in \mathbb{R}^n$ such that $\begin{pmatrix} A \\ -c^{\mathrm{T}} \end{pmatrix} v \le \begin{pmatrix} b \\ -\mu \end{pmatrix}$. We can apply Farkas's lemma, Lemma 29.3, to inequalty (29.49) with

$$M = \begin{pmatrix} A \\ -c^{\mathrm{T}} \end{pmatrix} \text{ and } g = \begin{pmatrix} b \\ -\mu \end{pmatrix} .$$

Because the augmented primal is infeasible, condition 1 of Farkas's lemma does not hold. Therefore, condition 2 must apply, so that there must exist a $w \in \mathbb{R}^{m+1}$ such that $w \ge 0$, $w^{\mathrm{T}}M = 0$, and $w^{\mathrm{T}}g < 0$. Let's write w as $w = \begin{pmatrix} \bar{y} \\ \lambda \end{pmatrix}$ for some $\bar{y} \in \mathbb{R}^m$ and $\lambda \in \mathbb{R}$, where $\bar{y} \geq 0$ and $\lambda \geq 0$. Substituting for w, M, and g in condition 2 gives

$$\begin{pmatrix} \bar{y} \\ \lambda \end{pmatrix}^{\mathrm{T}} \begin{pmatrix} A \\ -c^{\mathrm{T}} \end{pmatrix} = 0 \text{ and } \begin{pmatrix} \bar{y} \\ \lambda \end{pmatrix}^{\mathrm{T}} \begin{pmatrix} b \\ -\mu \end{pmatrix} < 0 .$$

Unpacking the matrix notation gives

$$\bar{y}^{\mathrm{T}} A - \lambda c^{\mathrm{T}} = 0 \text{ and } \bar{y}^{\mathrm{T}} b - \lambda \mu < 0 . \tag{29.50}$$

We now show that the requirements in (29.50) contradict the assumption that μ is the optimal solution value for the dual linear program. We consider two cases.

The first case is when $\lambda = 0$. In this case, (29.50) simplifies to

$$\bar{y}^{\mathrm{T}} A = 0 \text{ and } \bar{y}^{\mathrm{T}} b < 0 . \tag{29.51}$$

We'll now construct a dual feasible solution y' with an objective value smaller than $b^{\mathrm{T}} y^*$. Set $y' = y^* + \epsilon \bar{y}$, for any $\epsilon > 0$. Since

$$\begin{aligned} y'^{\mathrm{T}} A &= (y^* + \epsilon \bar{y})^{\mathrm{T}} A \\ &= y^{*\mathrm{T}} A + \epsilon \bar{y}^{\mathrm{T}} A \\ &= y^{*\mathrm{T}} A && \text{(by (29.51))} \\ &\geq c^{\mathrm{T}} && \text{(because } y^* \text{ is feasible),} \end{aligned}$$

y' is feasible. Now consider the objective value

$$\begin{aligned} b^{\mathrm{T}} y' &= b^{\mathrm{T}} (y^* + \epsilon \bar{y}) \\ &= b^{\mathrm{T}} y^* + \epsilon b^{\mathrm{T}} \bar{y} \\ &< b^{\mathrm{T}} y^*, \end{aligned}$$

where the last inequality follows because $\epsilon > 0$ and, by (29.51), $\bar{y}^{\mathrm{T}} b = b^{\mathrm{T}} \bar{y} < 0$ (since both $\bar{y}^{\mathrm{T}} b$ and $b^{\mathrm{T}} \bar{y}$ are the inner product of b and $\bar{y}$), and so their product is negative. Thus we have a feasible dual solution of value less than μ, which contradicts μ being the optimal objective value.

We now consider the second case, where $\lambda > 0$. In this case, we can take (29.50) and divide through by λ to obtain

$$(\bar{y}^{\mathrm{T}}/\lambda)A - (\lambda/\lambda)c^{\mathrm{T}} = 0 \text{ and } (\bar{y}^{\mathrm{T}}/\lambda)b - (\lambda/\lambda)\mu < 0 . \tag{29.52}$$

Now set $y' = \bar{y}/\lambda$ in (29.52), giving

$y'^{\mathrm{T}}A = c^{\mathrm{T}}$ and $y'^{\mathrm{T}}b < \mu$.

Thus, y' is a feasible dual solution with objective value strictly less than μ, a contradiction. We conclude that the augmented primal has a feasible solution, and the theorem is proved.

■

Fundamental theorem of linear programming

We conclude this chapter by stating the fundamental theorem of linear programming, which extends Theorem 29.4 to the cases when the linear program may be either feasible or unbounded. Exercise 29.3-8 asks you to provide the proof.

Theorem 29.5 (Fundamental theorem of linear programming)
Any linear program, given in standard form, either

1. has an optimal solution with a finite objective value,
2. is infeasible, or
3. is unbounded.

■

Exercises

29.3-1
Formulate the dual of the linear program given in lines (29.6)–(29.10) on page 852.

29.3-2
You have a linear program that is not in standard form. You could produce the dual by first converting it to standard form, and then

taking the dual. It would be more convenient, however, to produce the dual directly. Explain how to directly take the dual of an arbitrary linear program.

29.3-3

Write down the dual of the maximum-flow linear program, as given in lines (29.25)–(29.28) on page 862. Explain how to interpret this formulation as a minimum-cut problem.

29.3-4

Write down the dual of the minimum-cost-flow linear program, as given in lines (29.29)–(29.30) on page 864. Explain how to interpret this problem in terms of graphs and flows.

29.3-5

Show that the dual of the dual of a linear program is the primal linear program.

29.3-6

Which result from Chapter 24 can be interpreted as weak duality for the maximum-flow problem?

29.3-7

Consider the following 1-variable primal linear program:

$$
\begin{array}{ll}
\text{maximize} & tx \\
\text{subject to} & \\
 & rx \le s \\
 & x \ \ge 0 ,
\end{array}
$$

where r, s, and t are arbitrary real numbers. State for which values of r, s, and t you can assert that

1. Both the primal linear program and its dual have optimal solutions with finite objective values.
2. The primal is feasible, but the dual is infeasible.
3. The dual is feasible, but the primal is infeasible.

4. Neither the primal nor the dual is feasible.

29.3-8

Prove the fundamental theorem of linear programming, Theorem 29.5.

Problems

29-1 Linear-inequality feasibility

Given a set of m linear inequalities on n variables $x_1, x_2, \ldots, x_n$, the ***linear-inequality feasibility problem*** asks whether there is a setting of the variables that simultaneously satisfies each of the inequalities.

a. Given an algorithm for the linear-programming problem, show how to use it to solve a linear-inequality feasibility problem. The number of variables and constraints that you use in the linear-programming problem should be polynomial in n and m.

b. Given an algorithm for the linear-inequality feasibility problem, show how to use it to solve a linear-programming problem. The number of variables and linear inequalities that you use in the linear-inequality feasibility problem should be polynomial in n and m, the number of variables and constraints in the linear program.

29-2 Complementary slackness

Complementary slackness describes a relationship between the values of primal variables and dual constraints and between the values of dual variables and primal constraints. Let $\bar{x}$ be a feasible solution to the primal linear program given in (29.31)–(29.33), and let $\bar{y}$ be a feasible solution to the dual linear program given in (29.34)–(29.36). Complementary slackness states that the following conditions are necessary and sufficient for $\bar{x}$ and $\bar{y}$ to be optimal:

$$\sum_{i=1}^{m} a_{ij}\bar{y}_i = c_j \text{ or } \bar{x}_j = 0 \text{ for } j = 1, 2, \ldots, n$$

and